

# Robustness and Failure Threshold Analysis of Local Feature Detection Algorithms Under Image Degradation and Enhancement

Nazib Uddin Ahrar

*Dept. of Computer Science and Engineering  
BRAC University  
Dhaka, Bangladesh*

Maharin Sharif Abony

*Dept. of Computer Science and Engineering  
BRAC University  
Dhaka, Bangladesh*

**Abstract**—Local feature detection is a key component of many computer vision applications, including image matching, registration, tracking, and recognition. While classical detectors such as Harris, SIFT, ORB, and AKAZE perform well under moderate image transformations, their reliability can decline significantly when images are severely degraded. Most existing evaluation studies compare these detectors at a limited number of fixed degradation levels and provide general performance rankings, leaving their precise failure thresholds largely unexplored. This study introduces a controlled experimental framework to investigate how local feature detectors respond to progressively increasing levels of Gaussian noise, Gaussian blur, and reduced illumination. Using images from the HPatches dataset, four detectors are evaluated across 20 illumination and viewpoint sequences based on correct-match retention, matching accuracy, and geometric error. The results reveal distinct failure patterns among the detectors. SIFT reaches its failure threshold earliest under noise and blur, whereas AKAZE shows the greatest tolerance to both types of degradation. Harris, on the other hand, is particularly resilient to illumination loss but is the most sensitive to noise. The study also examines whether image enhancement can restore detector performance under degraded conditions. Three techniques—global histogram equalization, Contrast-Limited Adaptive Histogram Equalization (CLAHE), and gamma correction—are applied to the degraded images. The findings reveal a clear difference in how enhancement affects performance: it often leads to substantial recovery under reduced illumination, but offers little improvement or may even reduce performance under Gaussian noise and blur. These results provide quantitative, detector-specific failure thresholds and highlight the importance of selecting enhancement techniques based on the type of degradation, rather than treating them as a universal solution for improving feature detection robustness.

**Index Terms**—feature detection, SIFT, ORB, AKAZE, Harris, image degradation, robustness, failure threshold, image enhancement, CLAHE, HPatches

## I. INTRODUCTION

Local feature detection plays an important role in many computer vision applications, including image registration, visual tracking, object recognition, simultaneous localization and mapping (SLAM), and 3D reconstruction. The performance of these tasks largely depends on how consistently a detector can identify distinctive and repeatable keypoints under different imaging conditions [3], [10], [14]. Classical detectors like

Harris, SIFT, ORB, and AKAZE have demonstrated dependable performance under controlled conditions. Each of these detectors has unique benefits in corner detection, scale and rotation invariance, computational efficiency, and robustness to photometric changes [7], [15], [18]. However, several types of degradation frequently influence real-world photographs. Applications including mobile photography, autonomous navigation, and surveillance frequently experience motion or defocus blur, sensor noise, and low illumination. These conditions can significantly reduce detection performance, even when an algorithm is designed to handle such variations [8], [17].

Previous studies have compared feature detectors under different degradation conditions or examined their overall performance on benchmark datasets [9], [14], [19]. Although these studies provide useful insights into the strengths and limitations of different detectors, they generally do not identify the exact point at which a detector's performance becomes unreliable as degradation increases. This is an important distinction from a practical perspective. A detector that maintains its performance up to a certain noise level before experiencing a sharp decline may behave very differently from one whose performance gradually deteriorates. Similarly, image enhancement techniques such as contrast adjustment, CLAHE, and illumination correction are commonly used to improve degraded images. However, their impact on downstream feature detection, particularly their ability to delay or shift the point of failure, has not been thoroughly investigated [8], [11], [20].

To address these gaps, this study presents a controlled experimental framework for evaluating local feature detectors under progressively increasing levels of Gaussian noise, Gaussian blur, and reduced illumination. Images from the HPatches dataset are used to evaluate four representative detectors—Harris, SIFT, ORB, and AKAZE—across twenty illumination and viewpoint sequences. Their performance is assessed using several quantitative measures, including keypoint count, correct-match retention, matching accuracy, and geometric error. In addition to comparing their overall performance, this study identifies the degradation severity at which each detector falls below 50% retention of its clean-image baseline. The degraded images are also processed using three standard

enhancement techniques to examine whether preprocessing can improve feature detection and extend the reliable operating range of each detector.

The main contributions of this work are: (i) a quantitative characterization of the failure thresholds of individual feature detectors across three types of image degradation, and (ii) an empirical evaluation of image enhancement, identifying the conditions under which it improves, has little effect on, or negatively affects feature-matching performance.

## II. LITERATURE REVIEW

Local feature detection and matching are at the foundation of many computer vision applications such as image registration, localization, SLAM, object recognition, and 3D reconstruction. Their performance is affected by factors including repeatability, matching accuracy, localization quality, computational efficiency, and robustness to image degradation [3], [10], [14]. The available literature has shown that no single feature detector consistently performs well under all imaging conditions, which motivates a comparative evaluation under controlled degradation.

Harris corner detection provides a classical foundation for local feature extraction. Its simplicity and computational efficiency make it useful as a baseline for comparison [7]. However, studies have shown that Harris becomes less reliable under severe blur, scale changes, and other distortions [7], [12], which motivates its inclusion here as a reference point for how degradation affects a traditional, non-scale-invariant detector.

SIFT was developed to provide greater invariance to scale and rotation and is widely regarded as a strong benchmark for feature-matching accuracy [10], [18]. Prior studies report high correct-match counts and strong repeatability under scale, rotation, and geometric change [10], [12]. However, SIFT can become less reliable under significant illumination change and blur, where fewer useful features are detected [17]. Strong performance under clean or geometrically-transformed conditions therefore does not guarantee robustness under progressive degradation — a claim this paper tests directly.

ORB was introduced as a computationally efficient alternative to SIFT and SURF [15]. Its main advantages are low computational cost, rotation invariance, and suitability for real-time applications [6], [15]. However, ORB can sacrifice matching accuracy relative to SIFT under demanding conditions [2], [10], and recent work shows its keypoints may cluster in high-contrast regions, reducing spatial coverage even when raw match counts look favorable [4].

AKAZE and KAZE-based methods have received considerable attention for their behavior under challenging imaging conditions. Comparative studies indicate AKAZE offers a good balance between accuracy and computational cost and can perform particularly well under blur, photometric change, and geometric transformation [4], [9], [18], motivating its inclusion as a candidate expected to be comparatively degradation-tolerant.

Existing benchmarking studies commonly evaluate detectors using repeatability, correspondence count, matching accuracy, and runtime [3], [14], [18]. More recent work also considers the spatial distribution of keypoints, since a large raw match count is not useful if surviving features cluster in a small image region [4]. Work involving blur and noise further shows that detector ranking is not stable across degradation type — different studies report different best-performing algorithms depending on the degradation and the evaluation metric used [5], [9], [12]. This motivates evaluating robustness separately per degradation category rather than producing one composite ranking, which is the approach adopted in this study.

Low-light conditions are particularly challenging because contrast loss, noise, color distortion, and detail loss compound to damage feature extraction and matching [8], [20]. Several enhancement techniques have therefore been proposed to improve visibility and suppress noise in dark images [11], [20]. However, prior work also shows that enhancement does not necessarily improve feature matching simply because an image looks visually better: strong noise and illumination changes can still damage the local gradients and texture information that detectors rely on [8]. Modified extraction pipelines and enhancement-assisted methods have nonetheless shown improvements under difficult lighting [17], suggesting the benefit of enhancement may be degradation-type-dependent rather than universal — the central question this paper investigates empirically.

### A. Research Gap

Although prior studies compare Harris, SIFT, ORB, and AKAZE under different transformations, most evaluate performance at a small number of selected degradation levels or provide general rankings [9], [14], [19]. Less attention has been given to identifying the exact degradation level at which each detector becomes unreliable, and while enhancement methods have been shown to improve degraded and low-light images visually, their ability to shift or delay the failure point of local feature detectors has not been examined directly, per degradation type, per detector. This paper addresses both gaps using a progressive degradation sweep, a common 50%-retention failure criterion, and a post-hoc enhancement stage applied specifically at and around each detector's measured failure point.

## III. METHODOLOGY

This study adopts a controlled experimental framework to investigate the robustness and failure behaviour of local feature detection algorithms under progressively increasing image degradation. The primary objective is not only to compare detectors but to identify the degradation severity at which each becomes unreliable, and to test whether image enhancement can recover performance at that point.

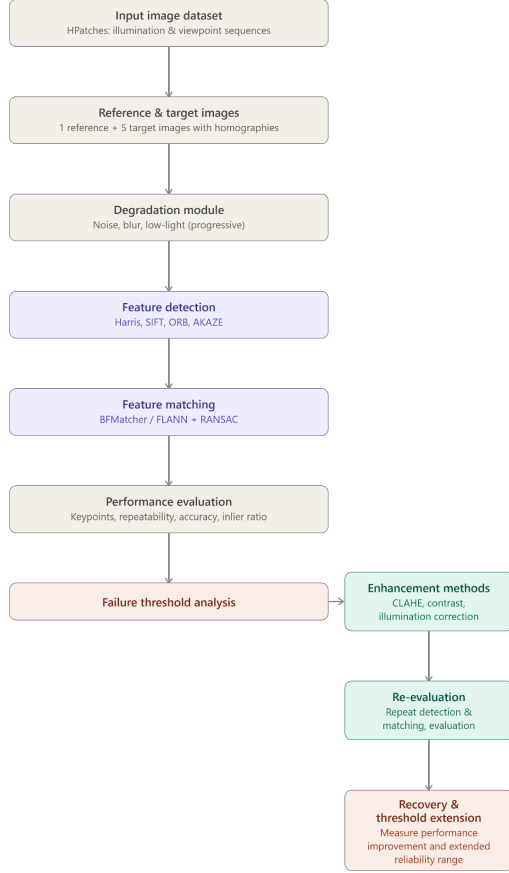


Fig. 1. Overview of the experimental pipeline: dataset selection, progressive degradation, feature detection and matching, failure threshold analysis, and the enhancement/re-evaluation stage.

### A. Dataset

The HPatches dataset was selected for this study because it provides a suitable benchmark for evaluating the robustness of local feature detectors under varying imaging conditions. Its illumination and viewpoint sequences allow the detectors to be tested on images with changes in brightness, contrast, and perspective. Additionally, the availability of ground-truth homography matrices makes it possible to establish reliable correspondences between reference and target images, supporting a consistent evaluation of keypoint detection and matching performance. A pilot subset of 20 sequences, consisting of 10 illumination and 10 viewpoint sequences, was used to examine detector behavior under image degradation and enhancement. This selection provides a balance between different types of visual variation while keeping the experimental setup manageable.

### B. Image Degradation Framework

Three degradation types were applied progressively to each target image:

- 1) **Gaussian noise** — additive noise with standard deviation  $\sigma \in \{0, 5, 10, 15, 20, 25, 30, 40, 50\}$ , generated with a fixed random seed per level for reproducibility and clipped to the valid 8-bit pixel range;
- 2) **Gaussian blur** — applied with kernel size auto-derived from  $\sigma \in \{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}$ ; and
- 3) **Illumination reduction** — simulated by linearly scaling pixel intensity by a brightness factor  $\in \{100\%, 90\%, \dots, 10\%\}$  of the original.

### C. Feature Detection and Matching Pipeline

Four feature detectors were evaluated: SIFT, ORB, AKAZE, and Harris. SIFT uses 128-D floating-point descriptors with L2 distance, while ORB was limited to 2000 features and uses 32-byte binary descriptors with Hamming distance. AKAZE used its default M-LDB descriptors with Hamming distance. For Harris, corners were detected using `goodFeaturesToTrack` with the Harris response and limited to 2000 points. SIFT descriptors were then computed at these Harris corner locations, allowing the Harris keypoints to be evaluated using the same descriptor-matching pipeline as SIFT.

For each detector, descriptors from the clean reference image were matched with those from the target image, which could have been degraded. A brute-force matcher with k-NN ( $k = 2$ ) was used, followed by Lowe’s ratio test with a threshold of 0.75 to filter out unreliable matches. The remaining matches were then checked geometrically using the ground-truth homography. Each reference keypoint was projected into the target image, and a match was considered correct when the Euclidean distance between the projected location and the matched target keypoint was no more than 5 pixels.

### D. Evaluation Metrics

For every (sequence  $\times$  detector  $\times$  degradation level) combination, four metrics were recorded: label=(iv)

- 1) number of detected keypoints;
- 2) number of ratio-test (“good”) matches;
- 3) number of geometrically correct matches;
- 4) matching accuracy (correct / good matches, %).

The primary robustness metric is correct-match retention, defined as the ratio of correct matches at a given degradation level to correct matches at the clean baseline, expressed as a percentage and pooled across all 20 sequences per detector per level. A failure threshold is defined as the first tested degradation level at which pooled retention drops below 50%.

### E. Enhancement Experiment

At and immediately around each detector’s measured failure level, three enhancement techniques were applied to the degraded image before re-running detection and matching: global histogram equalization, CLAHE (clip limit = 2.0,  $8 \times 8$  tile grid), and gamma correction ( $\gamma = 0.5$ , brightening). This produced 2,040 enhancement-condition records across the 20 sequences, four detectors, three degradation types,

and three enhancement methods, allowing a direct raw-vs-enhanced retention comparison at the stress levels nearest each detector’s failure point.

#### IV. RESULTS AND DISCUSSION

##### A. Clean-Image Baseline

On the single-sequence baseline pair (i\_ajuntament, identity homography), the four detectors produced markedly different keypoint counts and match quality, summarized in Table I.

TABLE I  
CLEAN-IMAGE BASELINE PERFORMANCE

| Detector | Target KP | Good | Correct | Accuracy | Med. Err (px) |
|----------|-----------|------|---------|----------|---------------|
| SIFT     | 780       | 348  | 286     | 82.18%   | 0.72          |
| ORB      | 1944      | 803  | 747     | 93.03%   | 0.00          |
| AKAZE    | 435       | 305  | 289     | 94.75%   | 0.49          |
| Harris   | 339       | 295  | 266     | 90.17%   | 1.00          |

Although ORB and AKAZE detected far fewer or far more raw keypoints than SIFT, both achieved higher geometric matching accuracy on clean images, confirming that keypoint count alone is a poor proxy for detector quality, consistent with the spatial-coverage concerns raised in prior work [4].

##### B. Robustness Under Gaussian Noise

Pooled correct-match retention across 20 sequences under increasing Gaussian noise is shown in Fig. 2. SIFT fails first, at  $\sigma = 20$  (49.82% retention). ORB and Harris fail later, at  $\sigma = 40$  (42.89%) and  $\sigma = 50$  (37.92%) respectively. AKAZE does not fail within the tested range, retaining 57.39% even at the maximum tested noise level. This ordering (AKAZE > Harris  $\approx$  ORB > SIFT under noise) contradicts the common assumption that SIFT is the most robust classical detector; here its scale-space extrema detection appears sensitive to additive noise, generating spurious low-response keypoints that survive detection but fail geometric verification.

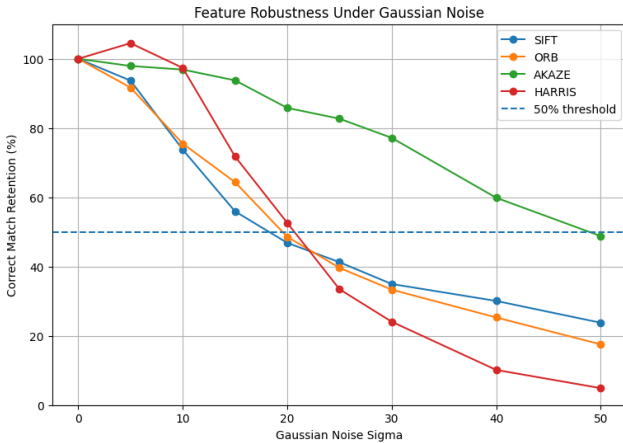


Fig. 2. Multi-sequence correct-match retention under increasing Gaussian noise ( $\sigma$ ). Dashed line marks the 50% failure criterion.

##### C. Robustness Under Gaussian Blur

Blur produces the steepest overall collapse of the three degradation types (Fig. 3); every detector drops from near-100% at  $\sigma = 0.5$  to below 90% by  $\sigma = 1.0$ . SIFT fails earliest ( $\sigma = 1.5$ , 39.17%), followed by ORB ( $\sigma = 2.0$ , 49.40%) and Harris ( $\sigma = 2.5$ , 30.61%), with AKAZE again the most tolerant ( $\sigma = 3.0$ , 42.09%). AKAZE’s nonlinear scale-space construction appears to preserve edge and blob structure under blur noticeably better than SIFT’s Gaussian-pyramid scale space, matching prior comparative findings [9], [18].

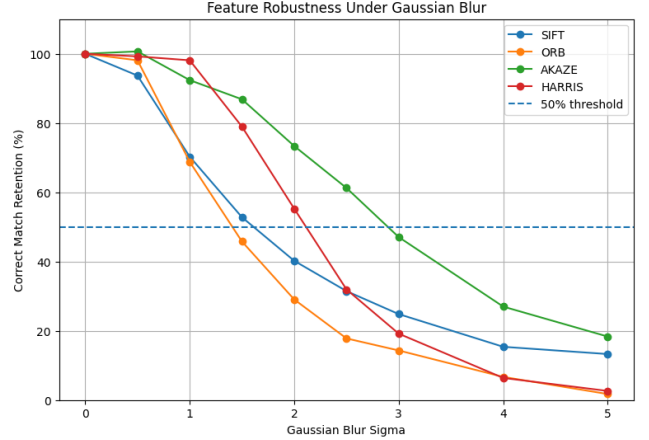


Fig. 3. Multi-sequence correct-match retention under increasing Gaussian blur ( $\sigma$ ).

##### D. Robustness Under Illumination Reduction

This degradation type produces the most striking result of the study (Fig. 4): Harris shows essentially no degradation across the entire illumination sweep, retaining  $\approx 100\%$  correct-match rate down to 10% brightness, and never crosses the 50% failure criterion. This is expected mechanistically — the Harris corner response and the SIFT descriptors computed at Harris-detected locations are built from image gradients, which are largely invariant to uniform multiplicative brightness scaling, whereas SIFT, ORB, and AKAZE’s own keypoint-detection stages depend on absolute intensity/contrast thresholds that are directly suppressed by dimming. Among the intensity-threshold-driven detectors, AKAZE fails earliest and most severely (50% brightness, 39.37% retention, collapsing to near 0% by 10% brightness) despite being the most noise- and blur-tolerant detector — illustrating that a detector’s ranking is not stable across degradation type [5], [9], [12]. SIFT fails at 40% brightness (40.91%), and ORB is the most tolerant of the three, failing only at 20% brightness (27.28%).

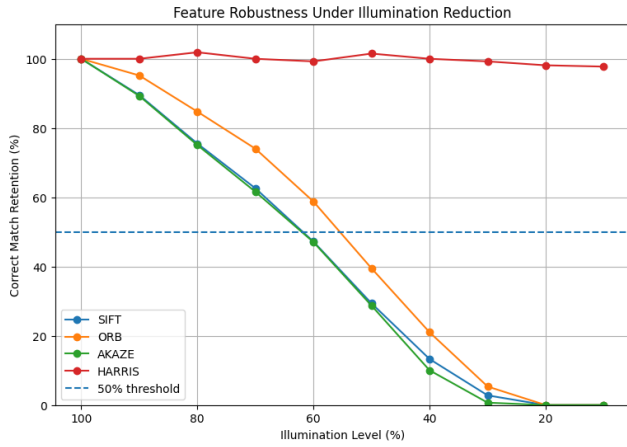


Fig. 4. Multi-sequence correct-match retention under progressive illumination reduction.

### E. Cross-Degradation Failure Threshold Summary

Table II summarizes the measured failure threshold for every detector and degradation type. No detector is uniformly best or worst: AKAZE is the strongest performer under noise and blur but the weakest under illumination loss; Harris is the weakest under noise but effectively degradation-proof under illumination loss. This confirms that single aggregate rankings obscure practically important, degradation-specific failure behavior, and that detector choice should be conditioned on the dominant degradation expected in deployment.

TABLE II  
FAILURE THRESHOLDS BY DETECTOR AND DEGRADATION TYPE

| Detector | Noise $\sigma$ | Blur $\sigma$ | Illum. % |
|----------|----------------|---------------|----------|
| SIFT     | 20             | 1.5           | 40       |
| ORB      | 40             | 2.0           | 20       |
| AKAZE    | none           | 3.0           | 50       |
| Harris   | 50             | 2.5           | none     |

### F. Effect of Enhancement at the Failure Point

Applying global histogram equalization, CLAHE, and gamma correction at each detector’s measured failure level produced a clear and consistent asymmetry between degradation types, visualized in Figs. 5–7 and summarized in Table III.

**Noise and blur.** (Figs. 5–6) enhancement did not recover pre-failure performance in almost every case. For SIFT at  $\sigma = 20$  noise, raw retention (49.82%) exceeded all three enhanced variants; the same pattern held for ORB and Harris under noise, and for SIFT, ORB, and Harris under blur. The sole exception was AKAZE under blur at  $\sigma = 3.0$ , where CLAHE raised retention from 42.09% to 55.07% (+12.98 points), successfully pushing it back above the 50% threshold. Histogram-based contrast operations amplify noise and sharpen blur artifacts along with genuine structure, which explains why they tend to worsen rather than help detectors

already stressed by noise or blur — consistent with prior findings that visual improvement does not always translate into recoverable structural detail for matching [8], [17].

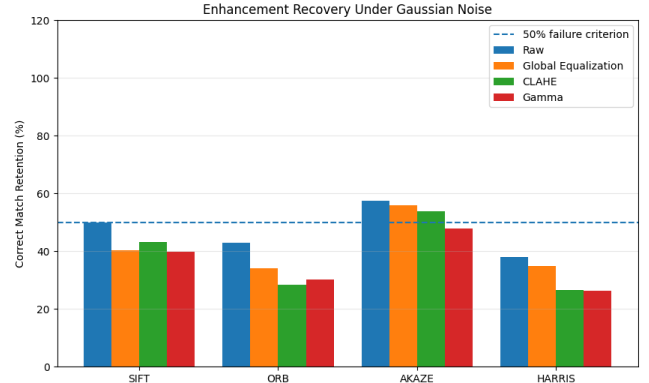


Fig. 5. Enhancement recovery under Gaussian noise, evaluated at each detector’s failure level.

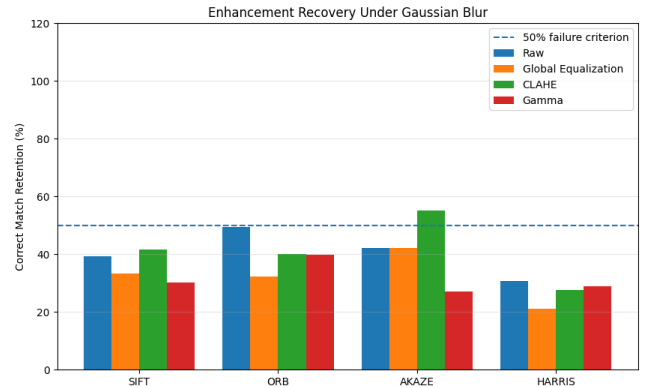


Fig. 6. Enhancement recovery under Gaussian blur, evaluated at each detector’s failure level.

**Illumination.** (Fig. 7) enhancement produced large, consistent gains. At SIFT’s failure level (40% brightness), CLAHE raised retention from 40.91% to 95.09% (+54.18 points) and Global Equalization to 72.34% (+31.43 points); both recovering well above the 50% threshold. AKAZE showed the same pattern (39.37% → 91.28% with CLAHE, +51.91 points), and ORB improved from 27.28% to 70.81% with CLAHE (+43.53 points). CLAHE was the most effective method in every illumination case tested, outperforming both global equalization and gamma correction, consistent with its design goal of locally adaptive contrast enhancement without the noise amplification characteristic of global equalization.

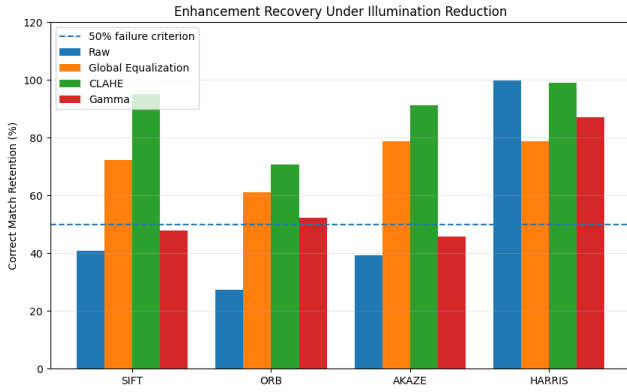


Fig. 7. Enhancement recovery under illumination reduction, evaluated at each detector’s failure level.

TABLE III  
RAW VS. BEST-ENHANCEMENT RETENTION AT FAILURE LEVELS

| Deg.  | Det.   | Lvl | Raw   | Best Enh.      | Gain (pp) |
|-------|--------|-----|-------|----------------|-----------|
| Noise | SIFT   | 20  | 49.82 | CLAHE (43.11)  | −6.70     |
| Noise | ORB    | 40  | 42.89 | Gl.Eq. (34.21) | −8.68     |
| Noise | AKAZE  | 50  | 57.39 | Gl.Eq. (55.79) | −1.60     |
| Noise | Harris | 50  | 37.92 | Gl.Eq. (34.75) | −3.17     |
| Blur  | SIFT   | 1.5 | 39.17 | CLAHE (41.54)  | +2.38     |
| Blur  | ORB    | 2.0 | 49.40 | CLAHE (40.18)  | −9.22     |
| Blur  | AKAZE  | 3.0 | 42.09 | CLAHE (55.07)  | +12.98    |
| Blur  | Harris | 2.5 | 30.61 | Gamma (28.97)  | −1.64     |
| Light | SIFT   | 40  | 40.91 | CLAHE (95.09)  | +54.18    |
| Light | ORB    | 20  | 27.28 | CLAHE (70.81)  | +43.53    |
| Light | AKAZE  | 50  | 39.37 | CLAHE (91.28)  | +51.91    |
| Light | Harris | 10  | 99.72 | CLAHE (99.04)  | −0.68     |

Of twelve tested (degradation × detector) failure conditions, enhancement recovered performance above the 50% threshold in four cases — all three intensity-threshold-driven detectors under illumination loss, plus AKAZE under blur — while noise-degraded conditions were never recovered by any enhancement method. This directly answers the study’s motivating question: enhancement is a targeted, illumination-specific mitigation rather than a general-purpose robustness improvement, and applying it indiscriminately to noisy or blurred imagery risks making feature matching worse, not better.

## V. CONCLUSION

This paper presented a controlled experimental framework for evaluating Harris, SIFT, ORB, and AKAZE under progressively increasing Gaussian noise, Gaussian blur, and illumination reduction, pooled across 20 HPatches sequences. Rather than offering only general performance rankings as in prior work [9], [14], [19], the proposed approach identifies the specific, quantitative degradation threshold at which each detector’s correct-match retention falls below a 50% usable level. The results show that no detector is uniformly robust: AKAZE tolerates noise and blur best but fails earliest under illumination loss, while Harris is essentially immune to illumination reduction but the most fragile of the four under noise,

and SIFT consistently fails earliest under both noise and blur despite its reputation as the classical robustness benchmark.

The study also tested whether contrast-enhancement techniques can delay these failure points, addressing prior evidence that visual improvement does not always translate into recoverable structural detail for matching [8], [17]. The results confirm this asymmetry sharply: enhancement — particularly CLAHE — reliably recovers and often dramatically improves detector performance under illumination-based degradation (gains up to +54 percentage points), but consistently fails to recover, and frequently worsens, performance under Gaussian noise and blur.

These findings offer practical, threshold-based guidance for applications such as low-light surveillance, outdoor robotics, and embedded vision, where imagery cannot be guaranteed clean: detector selection and enhancement strategy should be matched to the dominant expected degradation type rather than chosen from a single aggregate ranking. Future work could extend this comparison to learned/deep feature detectors, combined (multi-factor) degradation scenarios, and enhancement methods explicitly optimized for feature preservation rather than perceptual image quality.

## REFERENCES

- [1] M. Bansal, M. Kumar, and M. Kumar, “2D object recognition: a comparative analysis of SIFT, SURF and ORB feature descriptors,” *Multimedia Tools and Applications*, vol. 80, pp. 18839–18857, 2021.
- [2] E. Bayraktar and P. Boyraz, “Analysis of feature detector and descriptor combinations with a localization experiment for various performance metrics,” *arXiv preprint arXiv:1710.06232*, 2017.
- [3] M. Chekanov, O. Shipitko, and N. Skoryukina, “Study of keypoints detectors and descriptors performance on X-ray images compared to the visible light spectrum images,” *IEEE Access*, vol. 10, pp. 38964–38972, 2022.
- [4] A. Fesiuk and Y. Furgala, “Comprehensive spatial-geometric evaluation of keypoint detectors,” *Electronics and Information Technologies*, 2025.
- [5] M. Ghahremani, Y. Liu, and B. P. Tiddeman, “FFD: Fast feature detector,” *IEEE Transactions on Image Processing*, vol. 30, pp. 1153–1168, 2020.
- [6] O. Golovnin and D. V. Rybnikov, “Benchmarking of feature detectors and matchers using OpenCV-Python wrapper,” in *Proc. 2021 Int. Conf. on Information Technology and Nanotechnology (ITNT)*, 2021, pp. 1–6.
- [7] C. Harris and M. Stephens, “A combined corner and edge detector,” in *Proc. 4th Alvey Vision Conf.*, 1988, pp. 147–151.
- [8] Y. He, Y. Hu, W. Zhao, J. Li, Y. Liu, Y. Han, and J. Wen, “DarkFeat: Noise-robust feature detector and descriptor for extremely low-light RAW images,” in *Proc. AAAI Conf. Artificial Intelligence*, vol. 37, no. 1, 2023, pp. 826–834.
- [9] M. Işık, “Comprehensive empirical evaluation of feature extractors in computer vision,” *PeerJ Computer Science*, vol. 10, 2024.
- [10] S. Li, “A review of feature detection and match algorithms for localization and mapping,” *IOP Conf. Series: Materials Science and Engineering*, vol. 231, 2017.
- [11] H.-X. Liang, A.-S. Yu, M. Shao, and Y. Tian, “Multi-feature guided low-light image enhancement,” *Applied Sciences*, 2021.
- [12] C. Liu, J. Xu, and F. Wang, “A review of keypoints’ detection and feature description in image registration,” *Scientific Programming*, vol. 2021, art. 8509164, 2021.
- [13] M. Liu, Y. Cui, W. Ren, J. Zhou, and A. C. Knoll, “LIEDNet: A lightweight network for low-light enhancement and deblurring,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, pp. 6602–6615, 2025.
- [14] Y. Ou, Z. Cai, J. Lu, J. Dong, and Y. Ling, “Evaluation of image feature detection and matching algorithms,” in *Proc. 2020 5th Int. Conf. on Computer and Communication Systems (ICCCS)*, 2020, pp. 220–224.

- [15] E. Rublee, V. Rabaud, K. Konolige, and G. R. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *Proc. 2011 Int. Conf. on Computer Vision*, 2011, pp. 2564–2571.
- [16] E. S. Sabry, S. Elagooz, F. El-Samie, N. A. El-Bahnasawy, and G. M. El-Banby, "SIFT and ORB performance assessment for object identification in different test cases," *Journal of Optics*, vol. 53, pp. 1695–1708, 2023.
- [17] G. Tang, Z. Liu, and J. Xiong, "Distinctive image features from illumination and scale invariant keypoints," *Multimedia Tools and Applications*, pp. 1–28, 2019.
- [18] S. A. K. Tareen and Z. Saleem, "A comparative analysis of SIFT, SURF, KAZE, AKAZE, ORB, and BRISK," in *Proc. 2018 Int. Conf. on Computing, Mathematics and Engineering Technologies (iCoMET)*, 2018, pp. 1–10.
- [19] S. A. K. Tareen and R. H. Raza, "Potential of SIFT, SURF, KAZE, AKAZE, ORB, BRISK, AGAST, and 7 more algorithms for matching extremely variant image pairs," in *Proc. 2023 4th Int. Conf. on Computing, Mathematics and Engineering Technologies (iCoMET)*, 2023, pp. 1–6.
- [20] S. Yang, D. Zhou, J. Cao, and Y. Guo, "LightingNet: An integrated learning method for low-light image enhancement," *IEEE Transactions on Computational Imaging*, vol. 9, pp. 29–42, 2023.